

Tema 1: Números Complejos

a) $z_3 = \sqrt{2}e^{j\pi/2}$

b) $z_3 = 3.5 + j5.6$

c) $z_3 = 7.36 - j7.12$

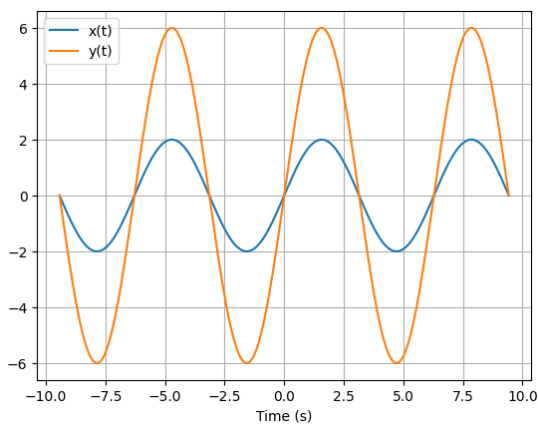
d) $z_3 = -3\sqrt{2}$

e) $z_3 = j10$

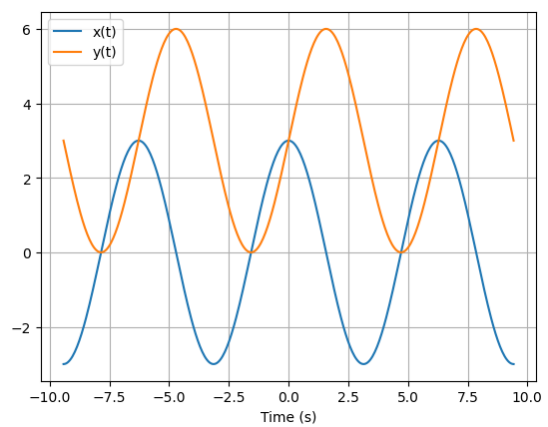
Tema 2: Operación de la variable dependiente/independiente

(1)

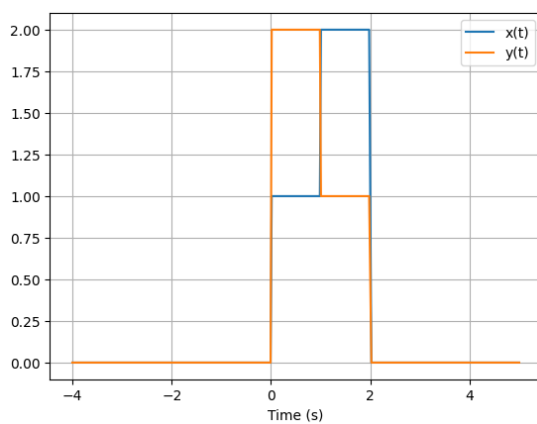
a)



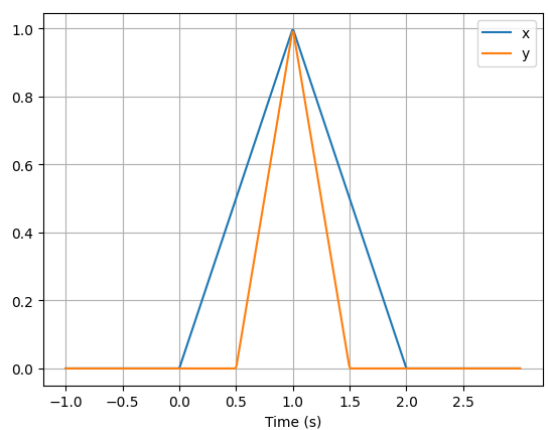
b)



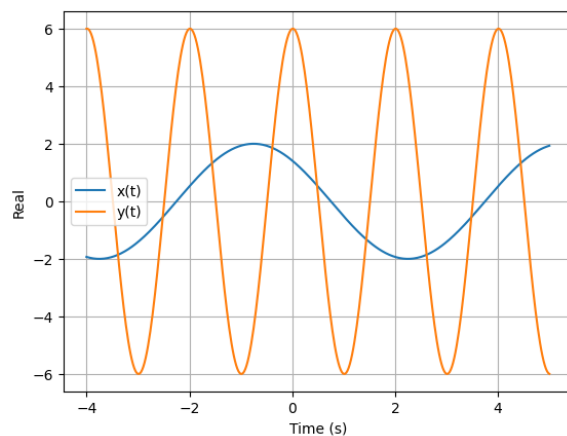
c)



d)

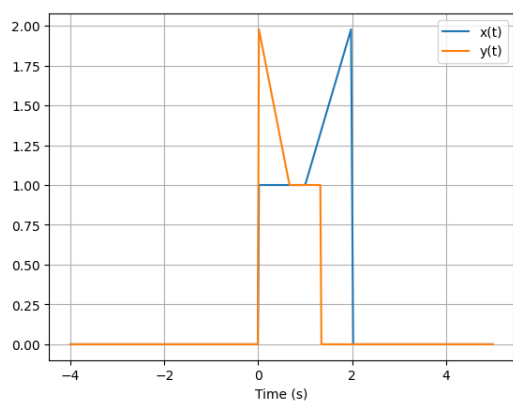


e)

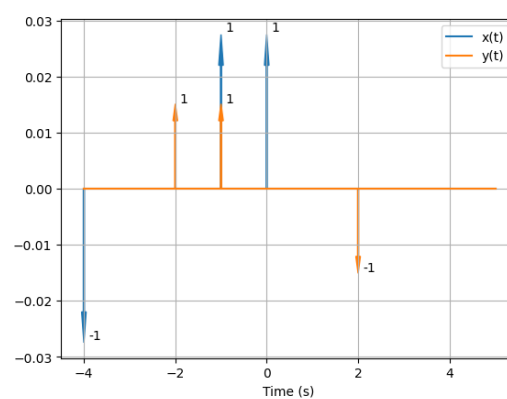


(2)

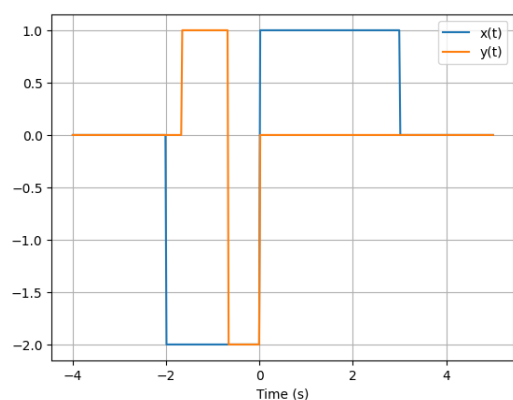
a) $x(t) = u(t) - u(t-1) + t * (u(t-1) - u(t-2))$



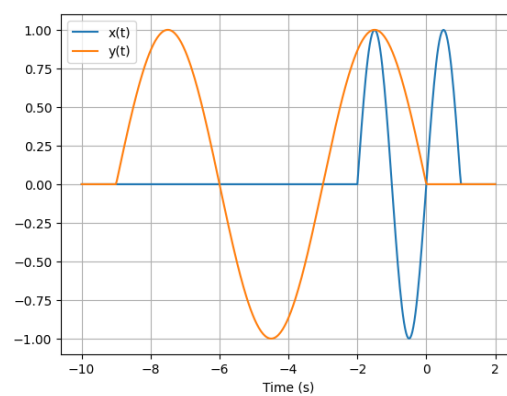
b) $x(t) = -\delta(t+4) + \delta(t+1) + \delta(t)$



c) $x(t) = -2u(t+2) + 3u(t) - u(t-3)$



d) $x(t) = \sin(\pi t) \cdot (u(t+2) - u(t-1))$

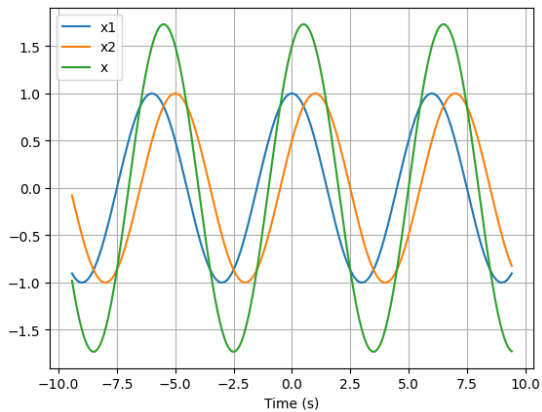


Tema 3: Periodicidad. Energía y Potencia de una señal

(1)

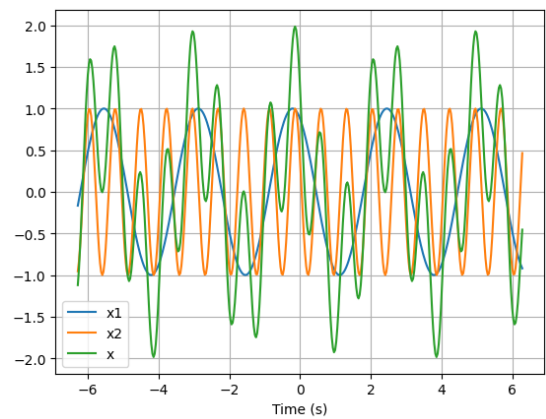
$$x(t) = \cos\left(\frac{\pi}{3}t\right) + \cos\left(\frac{\pi}{3}t - \pi/3\right)$$

$$T_1 = 6, T_2 = 6, T_0 = 6$$



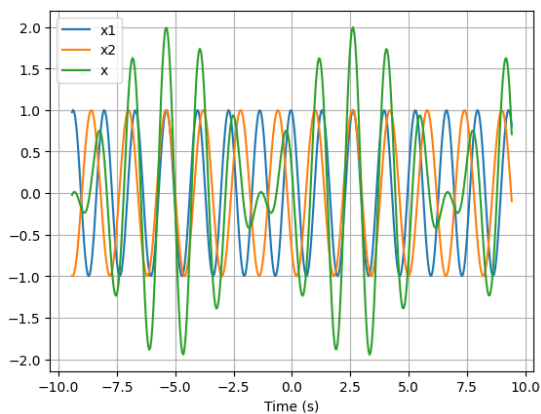
$$x(t) = \cos\left(\frac{3\pi}{4}t + \frac{1}{2}\right) + \cos\left(\frac{11\pi}{4}t + \frac{3\pi}{8}\right)$$

$$T_1 = 8/3, T_2 = 8/11, T_0 = 8$$



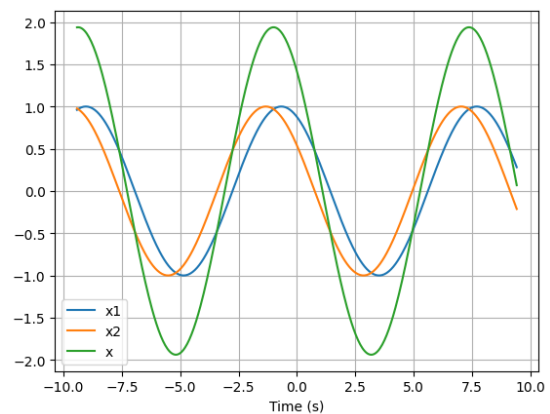
$$x(t) = \cos\left(\frac{3\pi}{2}t + \frac{1}{5}\right) + \cos\left(\frac{5\pi}{4}t + \frac{3\pi}{4}\right)$$

$$T_1 = 4/3, T_2 = 8/5, T_0 = 8$$

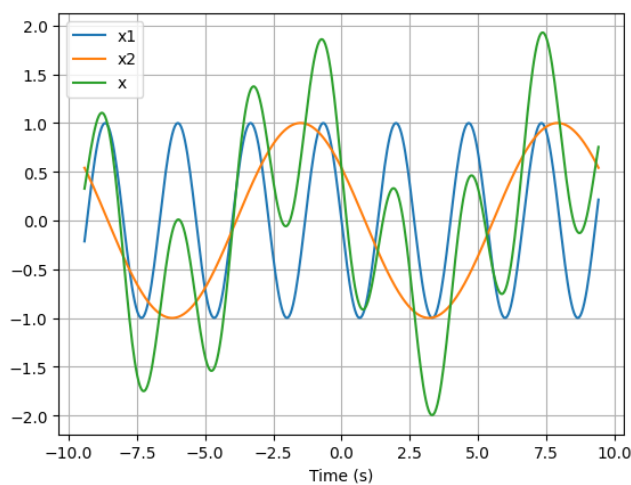


$$x(t) = \cos\left(\frac{3}{4}t + \frac{1}{2}\right) + \cos\left(\frac{3}{4}t + 1\right)$$

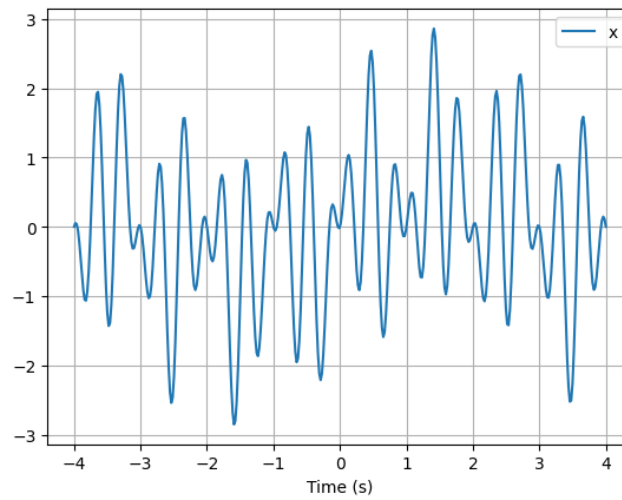
$$T_1 = 8\pi/3, T_2 = 8\pi/3, T_0 = 8\pi/3$$



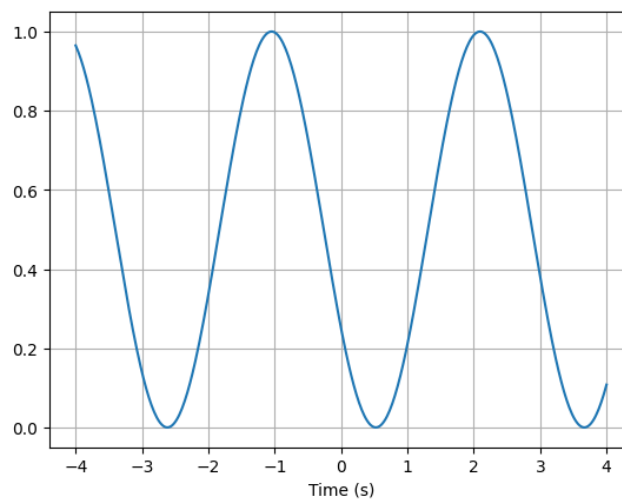
$$x(t) = \cos\left(\frac{3\pi}{4}t + \frac{\pi}{2}\right) + \cos\left(\frac{2}{3}t + 1\right), T_1 = 8/3, T_2 = 3\pi, \text{ NO ES PERIODICA}$$



b) $z(t) = \frac{1}{2}\text{sen}(\frac{\pi}{3}t) + \text{sen}(\frac{5\pi}{3}t) + \cos(\frac{13\pi}{3}t) - \cos(\frac{19\pi}{3}t)$, $T_0 = 6$



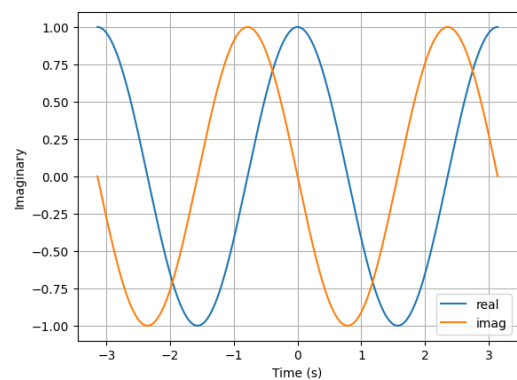
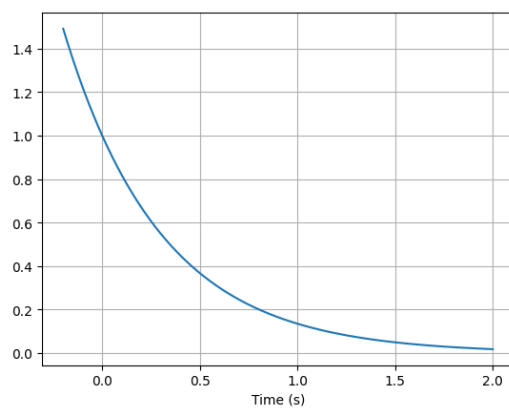
c) $x(t) = \frac{1}{2} \left[1 - \cos(2t - \frac{2}{6}\pi) \right]$, $T_0 = \pi$



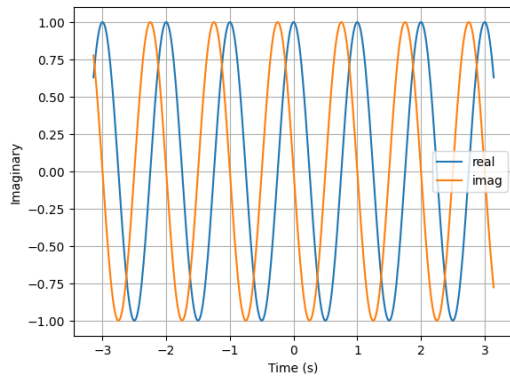
d) Determinar el período de las siguientes señales compuestas por exponenciales:

• $x(t) = e^{-2t}$, No es periódica

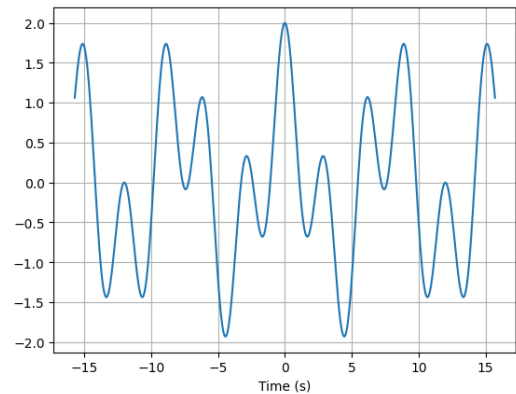
• $x(t) = e^{-2jt}$, $T_0 = \pi$



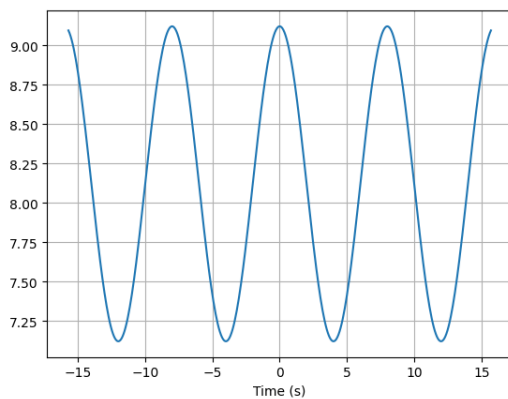
• $x(t) = e^{j2\pi t}, T_0 = 1$



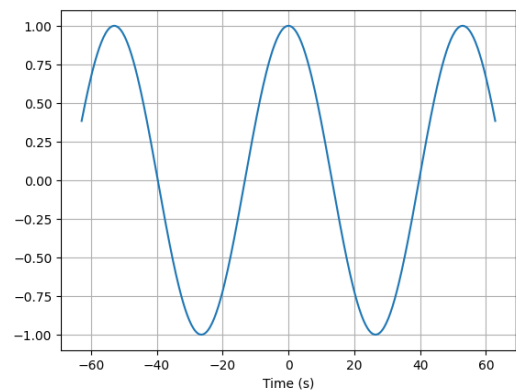
• $x(t) = e^{-j\pi t/4} + e^{j2\pi t/3}, T_0 = 24$



• $x(t) = e^{-j\pi t/4} + e^{2\pi/3}, T_0 = 8$



• $x(t) = e^{-j\pi t/4} \cdot e^{j2t/3}, T_0 = 52.92$



• $x(t) = e^{-j\pi t/4} + e^{j2t/3}, \text{ No es periódica}$

(2) Calcular la energía y potencia de la señal en un período (si es periódica) y en el infinito. Determinar si es una señal de energía.

a) $E_\infty = 16$

$P_{1;5} = 4$

$P_\infty = 0$

No es periódica. Es una señal de energía.

c) $E_\infty = \infty$

$P_\infty = \infty$

No es periódica, tampoco es una señal de energía.

e) $E_\infty = \frac{1}{4}$

$P_\infty = 0$

No es periódica, es una señal de energía.

b) $E_\infty = 64.67$

$P_{1;5} = 16.17$

$P_\infty = 0$

No es periódica. Es una señal de energía.

d) $E_\infty = \infty$

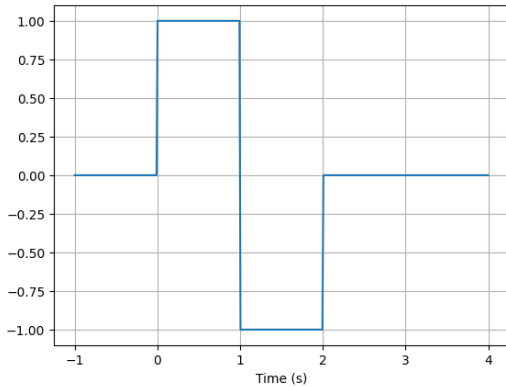
$P_\infty = \infty$

No es periódica, tampoco es una señal de energía.

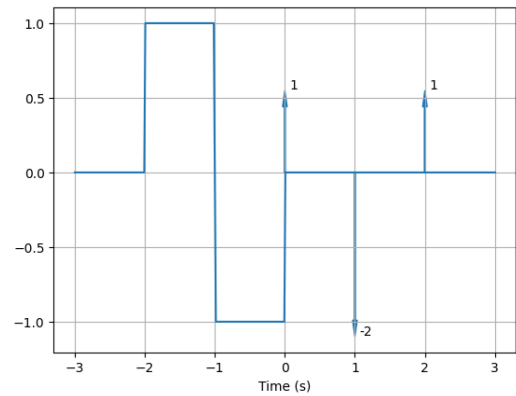
Tema 4: Derivadas e Integrales de una señal

(1) Calcular la derivada de las siguientes señales de manera analítica y gráfica.

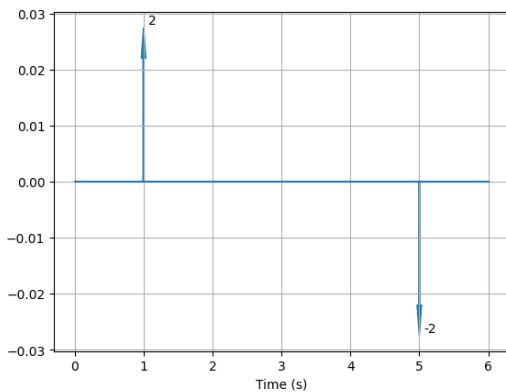
a) $x(t) = t \cdot [u(t) - u(t-1)] + (-t+2) \cdot [u(t-1) - u(t-2)]$
 $\frac{\partial}{\partial t} x(t) = u(t) - 2u(t-1) + u(t-2)$



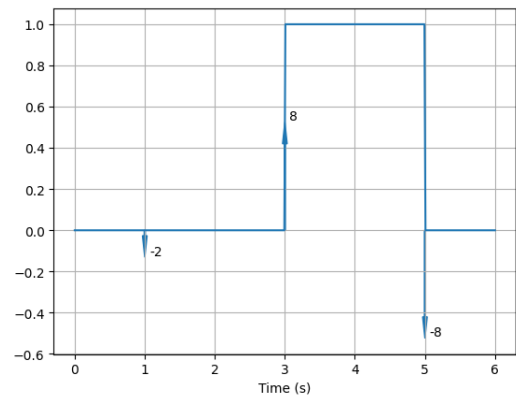
b) $x(t) = (t+2) \cdot (u(t+2) - u(t+1)) + (-t) \cdot (u(t+1) - u(t)) + u(t) - 2u(t-1) + u(t-2)$
 $\frac{\partial}{\partial t} x(t) = u(t+2) - 2u(t+1) + u(t) + \delta(t) - 2\delta(t-1) + \delta(t-2)$



c) $x(t) = 2 * u(t-1) - 2 * u(t-5)$
 $\frac{\partial}{\partial t} x(t) = 2 * \delta(t-1) - 2 * \delta(t-5)$

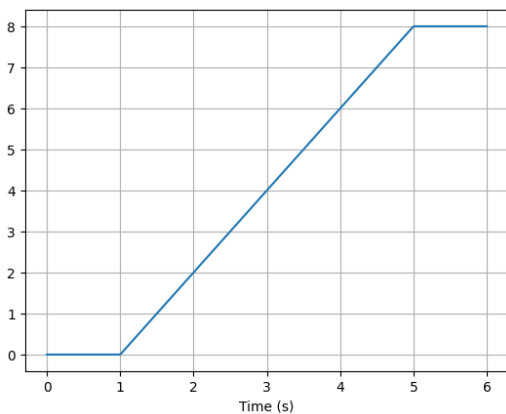


d) $x(t) = -2(u(t-1) - u(t-3)) + (t+3)(u(t-3) - u(t-5))$
 $\frac{\partial}{\partial t} x(t) = -2\delta(t-1) + 8\delta(t-3) + u(t-3) - u(t-5) - 8\delta(t-5)$

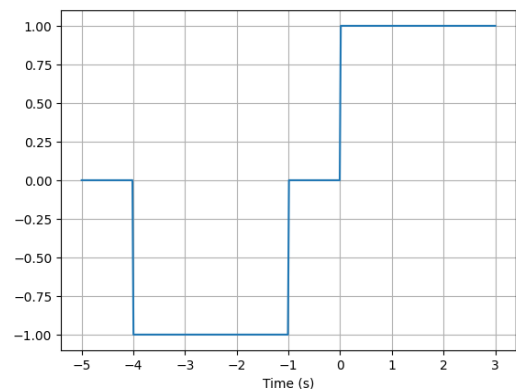


(2) Calcular la integral de las siguientes funciones

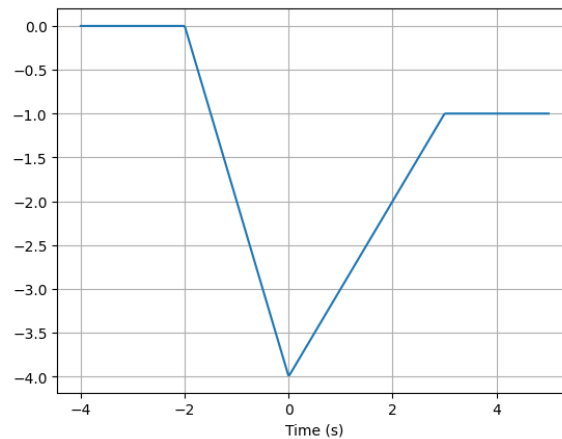
a) $x(t) = 2(u(t-1) - u(t-5))$
 $\int x(t)dt = (2t-2)u(t-1) - (2t-10)u(t-5)$



b) $x(t) = -\delta(t+4) + \delta(t+1) + \delta(t)$
 $\int x(t)dt = -u(t+4) + u(t+1) + u(t)$



$$c) \int x(t)dt = (-2t - 4)(u(t + 2) - u(t)) + (-4 + t) * (u(t) - u(t - 3)) - u(t - 3)$$



Tema 5: Sistemas

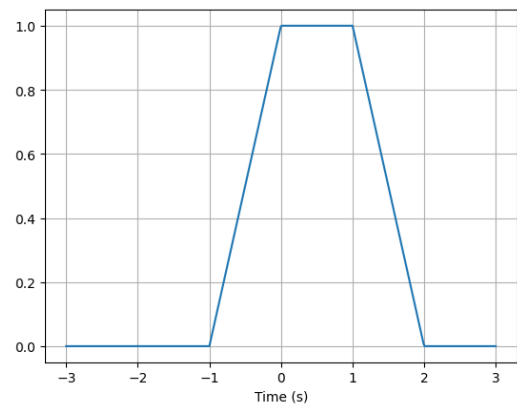
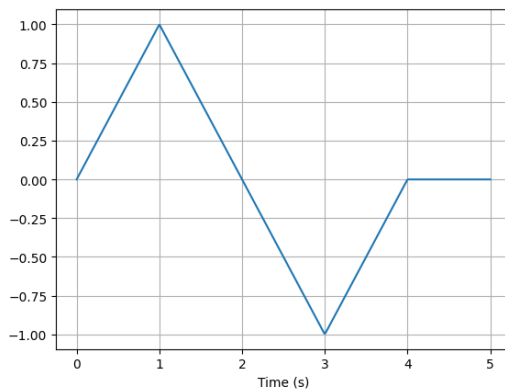
(1) Indicar si los sistemas expresados cumplen o no las propiedades de la tabla

$y(t)$	Memoria	Lineal	Invariante en el tiempo	Causal	Invertible	Estable
$(2 + \sin(t))x(t)$	No	Si	No	Si	No	Si
$x(2t)$	Si	Si	No	No	Si	Si
$\int_{-\infty}^t x(\tau)d\tau$	Si	Si	Si	Si	Si	No
$\frac{\partial x(t)}{\partial t}$	Si	Si	Depende	Si	No	No
$e^{x(t)}$	No	No	Si	Si	Si	Si
$x(t) \cdot x(t - 1)$	Si	No	Si	Si	No	Si
$\sin(6t) \cdot x(t)$	No	Si	No	Si	No	Si
$x(t - 1) - 2x(t - 4)$	Si	Si	Si	Si	Si	Si

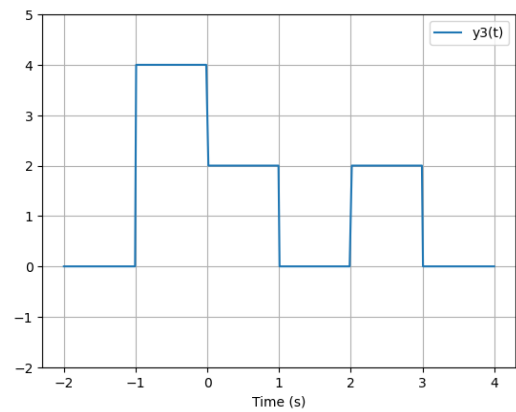
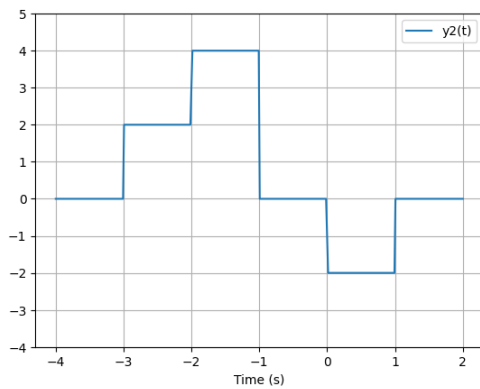
(2) Sea un sistema SLIT cuya respuesta a la señal $x_1(t)$ de la figura es la señal $y_1(t)$, determinar la respuestas de las entradas $x_2(t)$ y $x_3(t)$

a) $y_2 = t(u(t) - u(t - 1)) + (-t + 2)(u(t - 1) - u(t - 3)) + (t - 4)(u(t - 3) - u(t - 4))$

$y_3 = (t + 1)(u(t + 1) - u(t)) + (u(t) - u(t - 1)) + (-t + 2)(u(t - 1) - u(t - 2))$

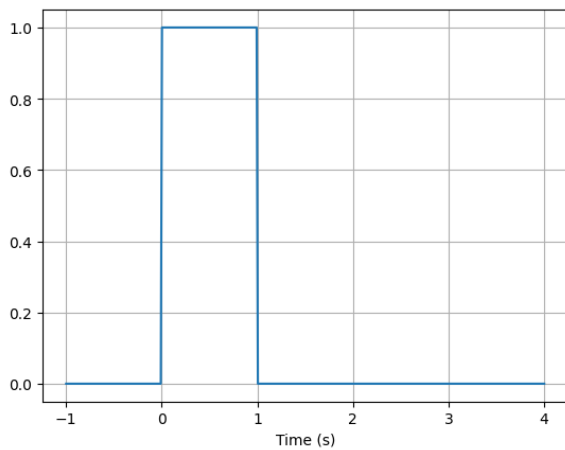


$$b) \quad y_2(t) = 2u(t+3) + 2u(t+2) - 4u(t+1) - 2u(t) + 2u(t-1) \quad y_3(t) = 4u(t+1) - 2u(t) - 2u(t-1) + 2u(t-2) - 2u(t-3)$$

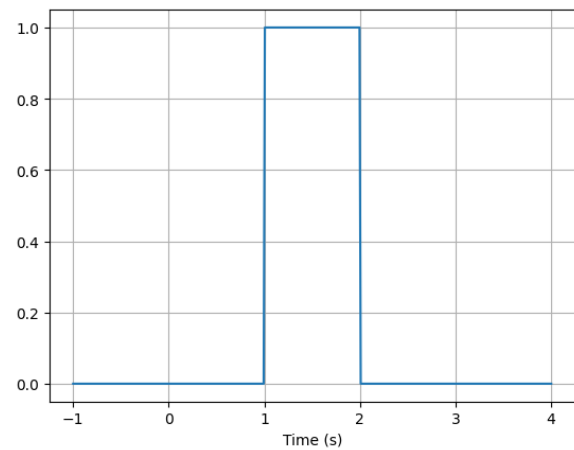


Tema 6: Convolución

(1)
a)



(a) $x(t)$



(b) $h(t)$

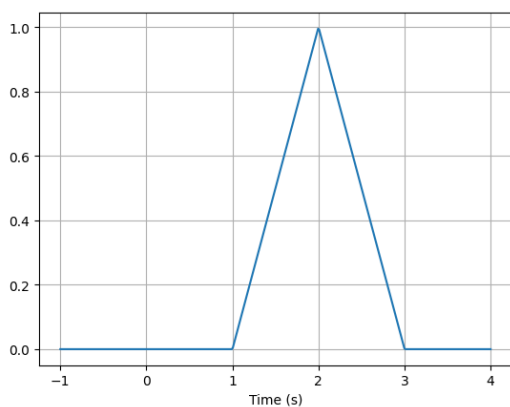
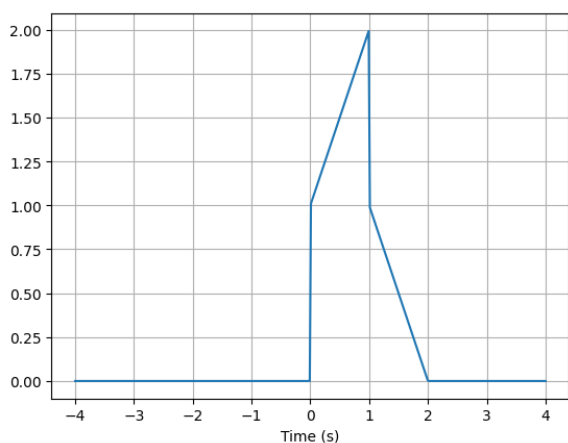


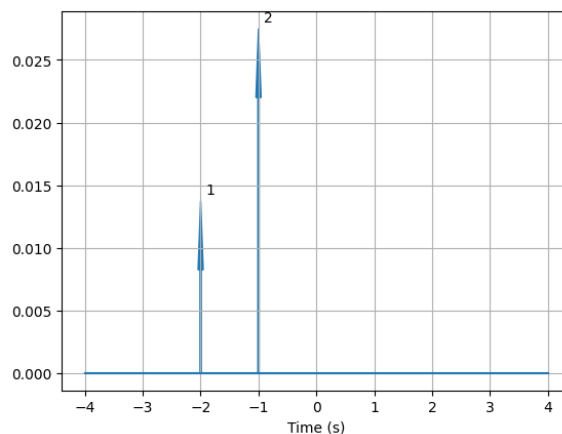
Figure 2: $y(t)$

$$y(t) = \begin{cases} 0, & t < 1 \\ t - 1, & 1 \leq t \leq 2 \\ -t + 3, & 2 \leq t \leq 3 \\ 0, & t > 3 \end{cases}$$

b)



(a) $x(t)$



(b) $h(t)$

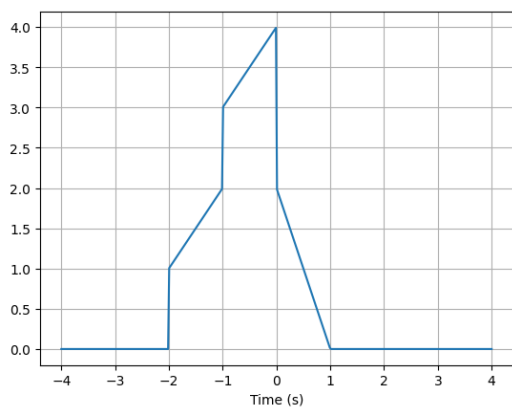
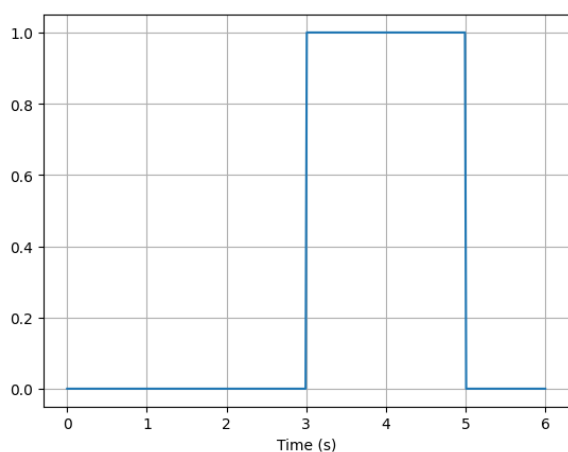


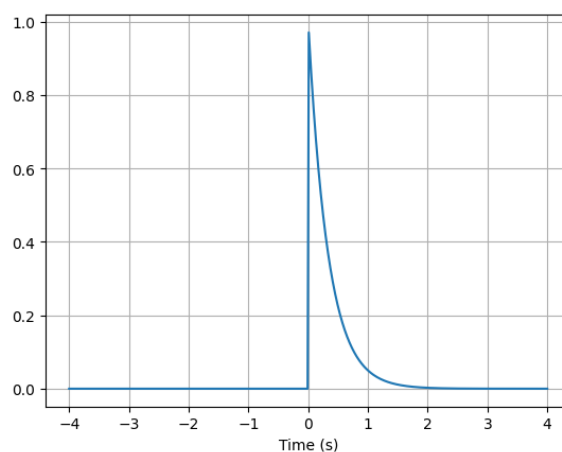
Figure 4: $y(t)$

$$y(t) = \begin{cases} 0 & t < -2 \\ t + 3 & -2 \leq t < -1 \\ t + 4 & -1 \leq t < 0 \\ -2t + 2 & 0 \leq t < 1 \\ 0 & t \geq 1 \end{cases}$$

c)



(a) $x(t)$



(b) $h(t)$

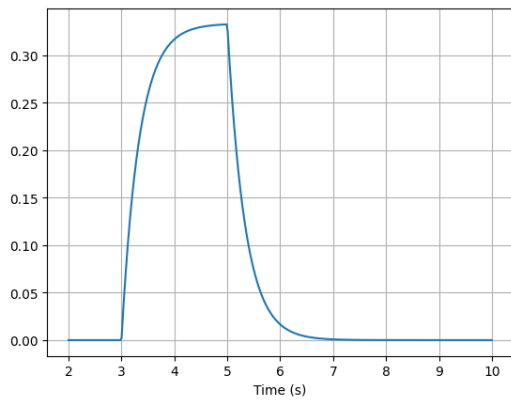
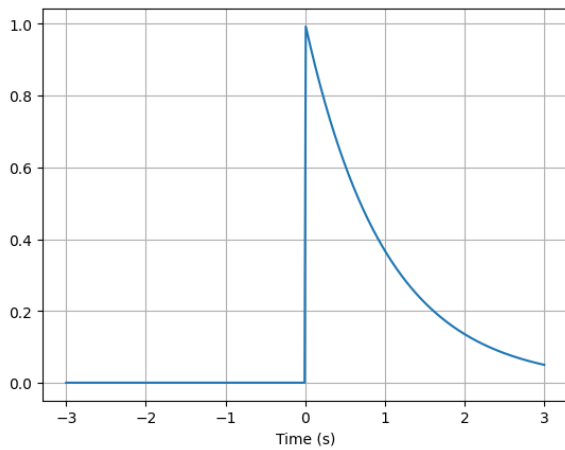


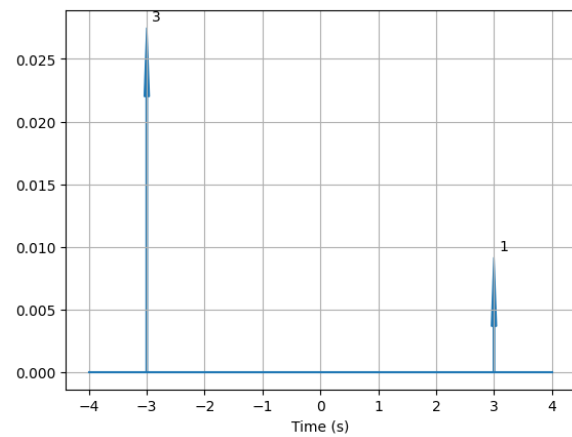
Figure 6: $y(t)$

$$y(t) = \begin{cases} 0 & t < 3 \\ -\frac{1}{3}e^{-3t+9} + \frac{1}{3} & 3 \leq t < 5 \\ -\frac{1}{3}e^{-3t}(e^9 - e^{15}) & 5 \leq t < \infty \end{cases}$$

d)



(a) $x(t)$



(b) $h(t)$

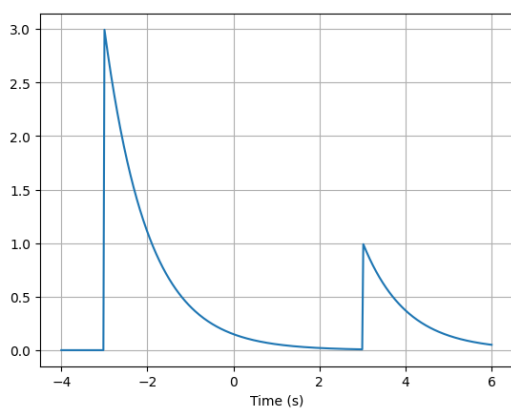
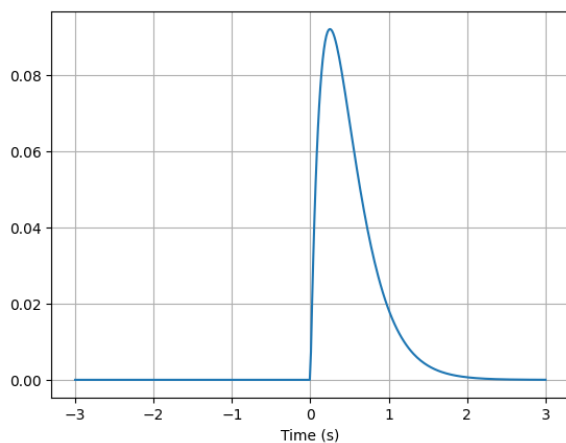


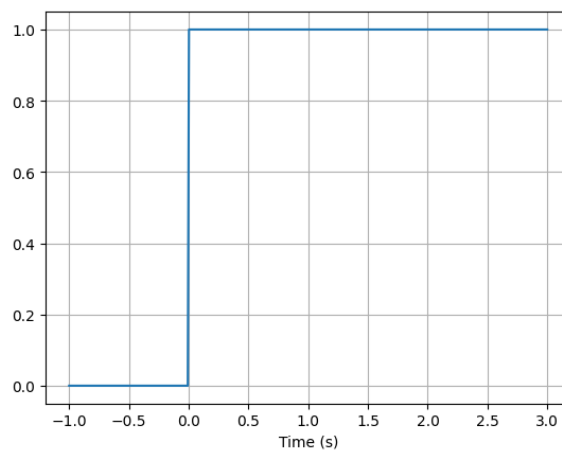
Figure 8: $y(t)$

$$y(t) = 3e^{-t-3}u(t+3) + e^{-t+3}u(t-3)$$

e)



(a) $x(t)$



(b) $h(t)$

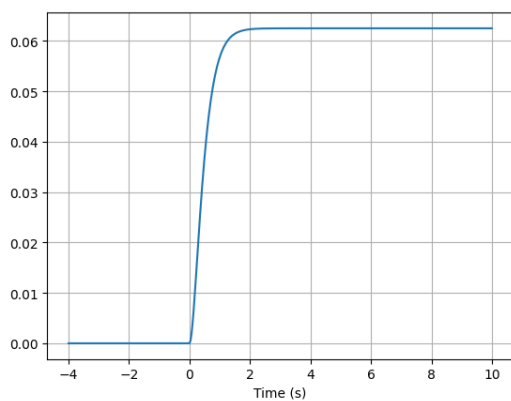
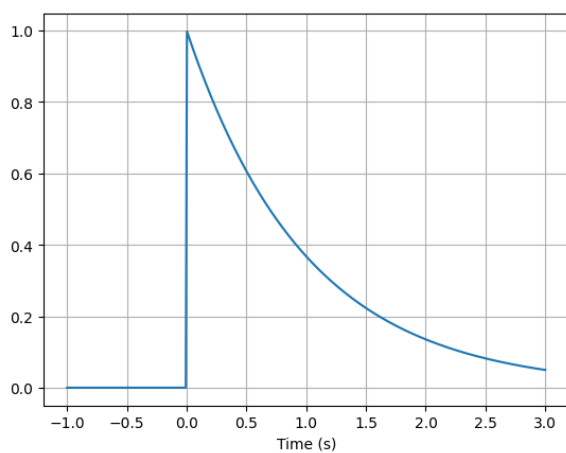


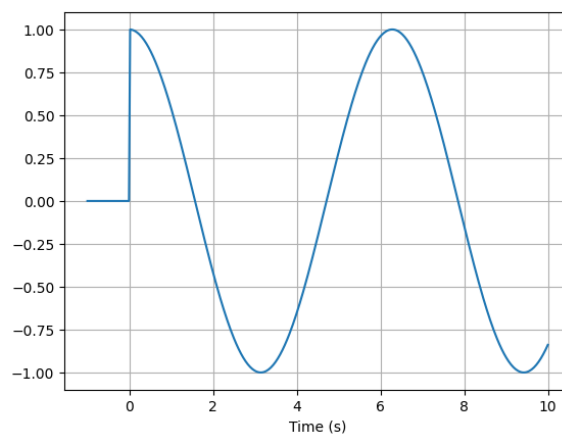
Figure 10: $y(t)$

$$y(t) = \frac{1 - e^{-\alpha t}(\alpha t + 1)}{\alpha^2}$$

f)



(a) $x(t)$



(b) $h(t)$

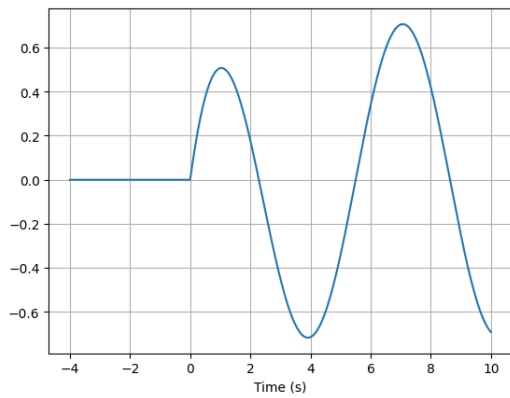
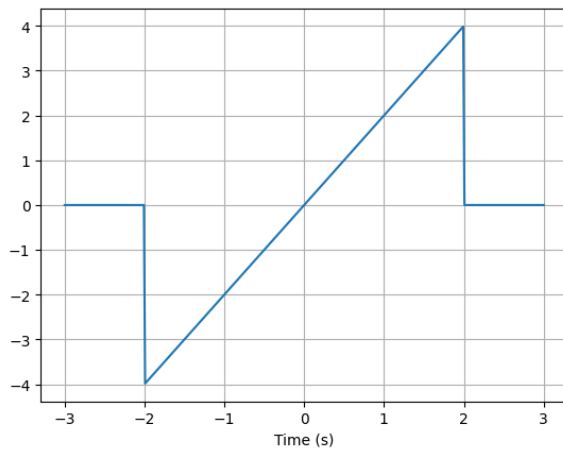


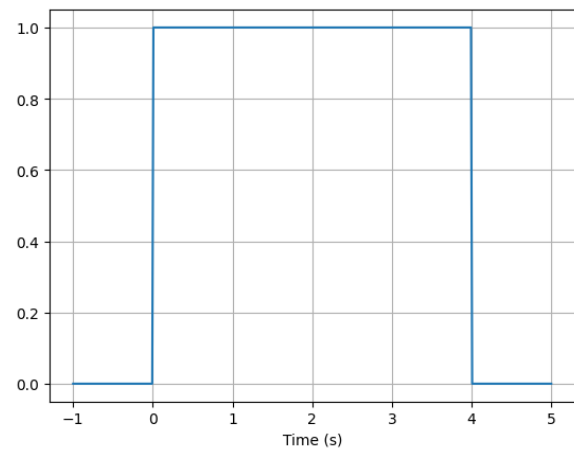
Figure 12: $y(t)$

$$y(t) = -\frac{1}{2}e^{-t} + \frac{1}{2}(\sin(t) + \cos(t))$$

g)



(a) $x(t)$



(b) $h(t)$

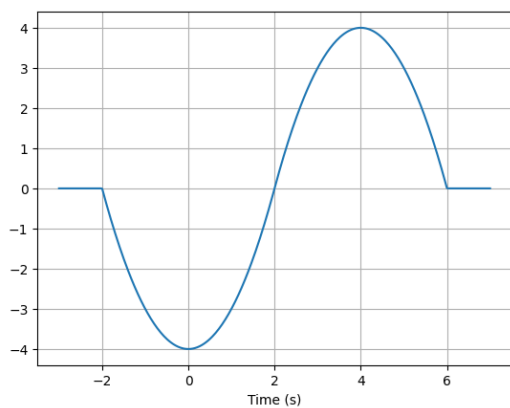


Figure 14: $y(t)$

$$y(t) = \begin{cases} 0 & t < -2 \\ t^2 - 4 & -2 \leq t < 2 \\ -t^2 + 8t - 12 & 2 \leq t < 6 \\ 0 & t \geq 6 \end{cases}$$

(2) Expresar las señales gráficas de forma matemática, determinar la convolución entre $x(t)$ y $h(t)$ y graficar el resultado:

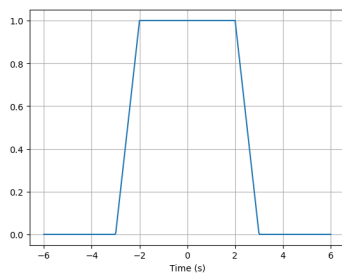
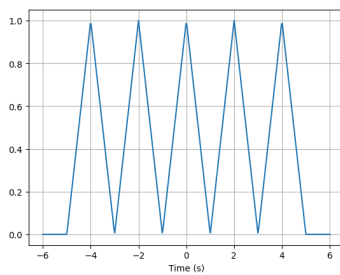
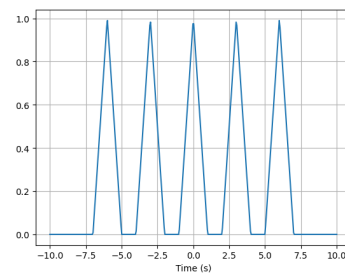
a)

$$x(t) = \sum_{n=-2}^2 \delta(t - nT)$$

$$h(t) = (t + 1)(u(t + 1) - u(t)) + (-t + 1)(u(t) - u(t - 1))$$

$$y(t) = \sum_{n=-2}^2 h(t - nT) = \sum_{n=-2}^2 (t - nT + 1)(u(t - nT + 1) - u(t - nT)) + (-t + nT + 1)(u(t - nT) - u(t - nT - 1))$$

Por ejemplo:

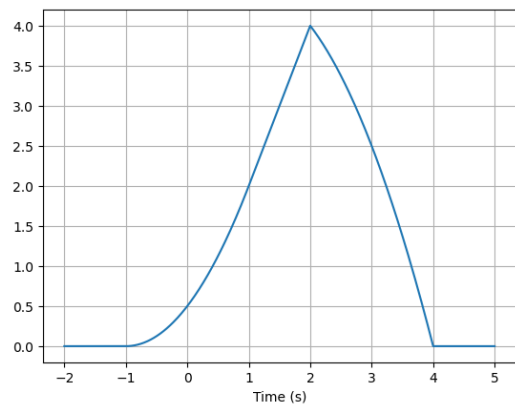

 (a) $y(t)$ para $T = 1$

 (b) $y(t)$ para $T = 2$

 (c) $y(t)$ para $T = 3$

b)

$$x(t) = u(t + 1) - u(t - 1)$$

$$h(t) = t * (u(t) - u(t - 3))$$

$$y(t) = \begin{cases} 0 & t < -1 \\ \frac{t^2}{2} + t + \frac{1}{2} & -1 \leq t < 1 \\ 2t & 1 \leq t < 2 \\ -\frac{t^2}{2} + t + 4 & 2 \leq t < 4 \\ 0 & t \geq 4 \end{cases}$$

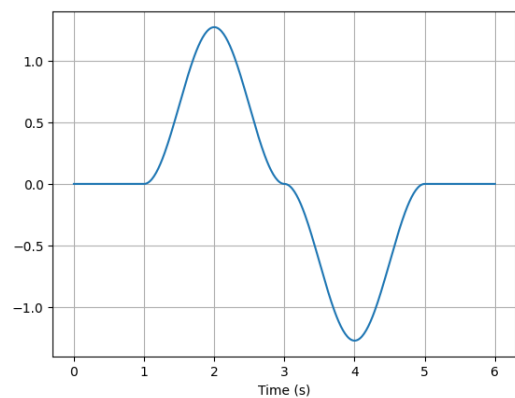

 Figure 16: $y(t)$

c)

$$x(t) = \sin(\pi t)[u(t) - u(t - 2)]$$

$$h(t) = 2[u(t - 1) - u(t - 3)]$$

$$y(t) = \begin{cases} 0 & t < 1 \\ -\frac{2}{\pi}[\cos(\pi(t - 1)) - 1] & 1 \leq t < 3 \\ \frac{2}{\pi}[\cos(\pi(t - 3)) - 1] & 3 \leq t < 5 \\ 0 & t \geq 5 \end{cases}$$

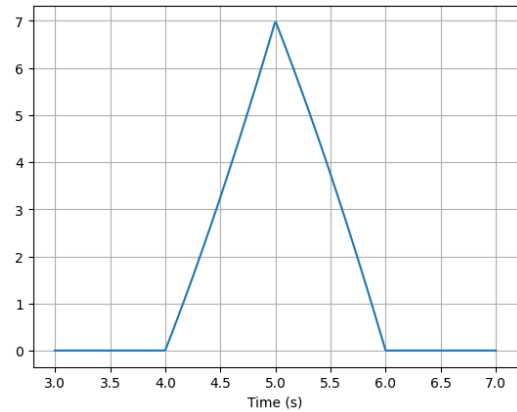

 Figure 17: $y(t)$

d)

$$x(t) = 2[u(t-2) - u(t-3)]$$

$$h(t) = (2t+2)(u(t-2) - u(t-3))$$

$$y(t) = \begin{cases} 0 & t < 4 \\ 2t^2 - 4t - 16 & 4 \leq t < 5 \\ -2t^2 + 8t + 24 & 5 \leq t < 6 \\ 0 & t \geq 6 \end{cases}$$

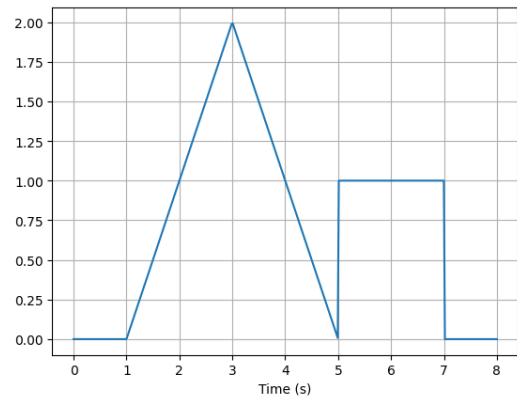

 Figure 18: $y(t)$

e)

$$x(t) = u(t+1) - u(t-1) + \delta(t-3)$$

$$h(t) = u(t-2) - u(t-4)$$

$$y(t) = \begin{cases} 0 & t < 1 \\ t-3 & 1 \leq t < 3 \\ -t+5 & 3 \leq t < 5 \\ 1 & 5 \leq t < 7 \\ 0 & t \geq 7 \end{cases}$$


 Figure 19: $y(t)$

(3) $A = t - 4$, $B = t - 5$

(4)

a)

$$y(t) = \begin{cases} t & 0 \leq t \leq \alpha \\ \alpha & \alpha \leq t \leq 1 \\ 1 + \alpha - t & 1 \leq t \leq (1 + \alpha) \\ 0 & \text{todo otro } t \end{cases}$$

 b) $\alpha = 1$

(5)

a)

$$y(t) = \begin{cases} 0 & t \leq 3 \\ \frac{1-e^{-3(t-3)}}{3} & 3 \leq t \leq 5 \\ \frac{(1-e^{-6})e^{-3(t-5)}}{3} & t > 5 \end{cases}$$

$$b) y_1(t) = e^{-3(t-3)}u(t-3) - e^{-3(t-5)}u(t-5)$$

Tema 7: Características de los sistemas según su respuesta al impulso

(1) Describir en cada sistema, según su respuesta al impulso $h(t)$, si tiene memoria, si es estable, si es causal. Además determinar la respuesta del sistema para la señal $x(t)$ y expresarlo de forma matemática y gráfica.

a)

$$x(t) = \delta(t-3)$$

$$h(t) = u(t) - u(t-2)$$

El sistema tiene memoria, es estable y es causal.

$$y(t) = u(t-3) - u(t-5)$$

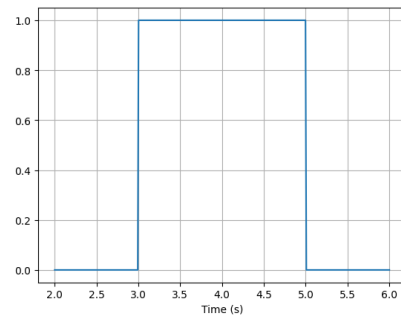


Figure 20: $y(t)$

b)

$$x(t) = u(t-3) - u(t-5)$$

$$h(t) = u(t)$$

El sistema tiene memoria, no es estable y es causal.

$$y(t) = \begin{cases} 0 & t < 3 \\ t-3 & 3 \leq t < 5 \\ 2 & t \geq 5 \end{cases}$$

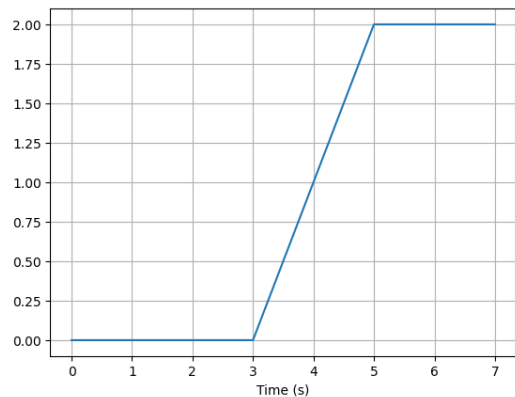


Figure 21: $y(t)$

c)

$$x(t) = -t(u(t+1) - u(t)) + t(u(t) - u(t-1)) + u(t-1) - u(t-2)$$

$$h(t) = u(t) - u(t-1)$$

El sistema tiene memoria, es estable y causal.

$$y(t) = \begin{cases} 0 & t < -1 \\ -\frac{t^2}{2} + \frac{1}{2} & -1 \leq t < 0 \\ t^2 - t + \frac{1}{2} & 0 \leq t < 1 \\ -\frac{t^2}{2} + 2t - 1 & 1 \leq t < 2 \\ -t + 3 & 2 \leq t < 3 \\ 0 & t \geq 3 \end{cases}$$

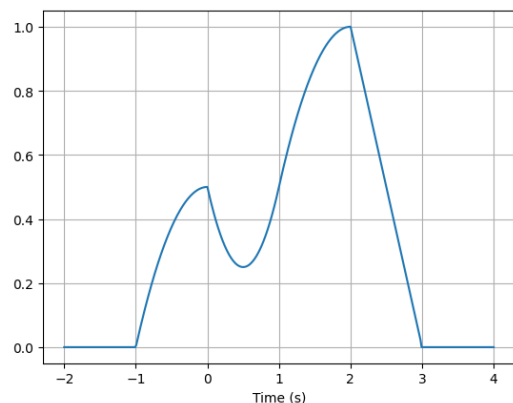


Figure 22: $y(t)$

Tema 8: Serie de Fourier

(1)

a)

$$a_1 = \frac{1}{2j}, \quad a_{-1} = -\frac{1}{2j}$$

b)

$$a_0 = 2, \quad a_2 = a_{-2} = -\frac{1}{2} \\ a_5 = -j2, \quad a_{-5} = j2$$

c)

$$a_k = -j3 \frac{\text{sen}^2(k\pi/2)}{k\pi}$$

d)

$$a_4 = a_{-4} = a_7 = a_{-7} = \frac{1}{2}$$

e)

$$a_0 = 1, \quad a_1 = a_{-1} = \frac{1}{4} \\ a_2 = a_{-2} = \frac{1}{2}, \quad a_3 = a_{-3} = \frac{1}{3}$$

f)

$$a_k = \frac{\text{sen}(kT_1 \frac{2\pi}{T})}{k\pi}$$

g)

$$a_1 = \frac{1}{2} + \frac{1}{2j}e^{j\pi/8} \\ a_{-1} = \frac{1}{2} - \frac{1}{2j}e^{-j\pi/8}$$

h)

$$a_1 = \frac{1}{2}e^{j\pi/6}, \quad a_{-1} = \frac{1}{2}e^{-j\pi/6} \\ a_2 = \frac{1}{2j}e^{-j\pi/3}, \quad a_{-2} = -\frac{1}{2j}e^{j\pi/3}$$

i)

$$a_1 = \frac{e^{j\pi/4}}{2j}, \quad a_{-1} = \frac{-e^{-j\pi/4}}{2j} \\ a_{-2} = a_0 = a_2 = \frac{1}{2}, \quad a_k = \frac{1}{2} \quad \forall k \in \text{par}$$

j)

$$a_2 = \frac{e^{j\pi/2}}{2} + \frac{1}{2}, \quad a_{-2} = \frac{e^{-j\pi/2}}{2} + \frac{1}{2} \\ a_{-1} = a_0 = a_1 = \frac{1}{2}, \quad a_k = \frac{1}{2}$$

Tema 9: Transformada de Fourier

(1) a)

$$X(\omega) = \frac{2\text{sen}(2\omega)}{\omega}e^{-j4\omega}$$

b)

$$X(\omega) = \frac{12}{36 + \omega^2}$$

c)

$$X(\omega) = 20\pi u(-\omega)e^{5\omega}$$

d)

$$X(\omega) = \frac{1}{3}\pi e^{-3|\omega|}$$

e)

$$X(\omega) = \frac{1}{8 + (\omega - 4)j}$$

f)

$$X(\omega) = \frac{2\text{sen}(2(\omega - 3))}{\omega - 3}$$

g)

$$X(\omega) = \frac{70}{49 + \omega^2}e^{j2\omega}$$

h)

$$X(\omega) = \sqrt{\pi} \frac{2 - \omega^2}{4} e^{\frac{-\omega^2}{4}}$$

i)

$$X(\omega) = \frac{e^{-2(6+j\omega)}}{(6 + j\omega)^2}$$

j)

$$X(\omega) = \frac{1}{j\omega} \left[-6\cos(3\omega) + \frac{2\text{sen}(3\omega)}{\omega} \right] + 4\pi\delta(\omega)$$

(2)

$$\begin{array}{ll} \text{a)} & y(t) = \frac{1}{12}e^{-6|t|} \\ \text{b)} & y(t) = \frac{5}{96}e^{2t}\mu(t) - \frac{5}{96}e^{-2t}\mu(t) - \frac{1}{96}e^{-10t}\mu(t) + \frac{1}{96}e^{10t}\mu(t) \\ \text{c)} & y(t) = -2.05e^{3t}\mu(t) - 0.18e^{-3t}\mu(t) + 0.36e^{-1.24t}\mu(t) - 1.87e^{3.23t}\mu(t) \\ \text{d)} & y(t) = -\frac{1}{48}e^{-t}\mu(t) - \frac{1}{48}e^t\mu(t) - \frac{1}{24}e^{-2t}\mu(t) \end{array}$$

(3)

$$\begin{array}{ll} \text{a)} & y(t) = 4\pi\cos(t/2) + 4\pi \\ \text{b)} & y(t) = 2\pi\cos(t) + 4\pi\cos(t/2) \end{array}$$

Tema 10: Serie/Transformada de Fourier

(1)

$$\begin{array}{ll} \text{a)} & a_k = \frac{10\text{sen}^2(k\pi/5)}{(k\pi)^2} \\ & a_0 = \frac{2}{5}, T = 5, \omega_0 = \frac{2\pi}{5} \\ \text{b)} & a_k = \frac{-j}{k\pi} \left[\frac{5\text{sen}(k4\pi/5)}{4k\pi} - \cos(k4\pi/5) \right] \\ & a_0 = 0, T = 5, \omega_0 = 2\pi/5 \\ \text{c)} & a_k = e^{-jk6\pi/7} \left[\frac{2\text{sen}(k2\pi/7)}{k\pi} + \frac{14\text{sen}^2(k\pi/7)}{(k\pi)^2} \right] \\ & a_0 = 6/7, T = 7, \omega_0 = 2\pi/7 \\ \text{d)} & a_k = \frac{j}{2} \frac{\text{sen}(k\pi/4)}{k\pi} \text{sen}(k\pi/2) e^{-j\pi k} \\ & a_0 = 0, T = 8, \omega_0 = \pi/4 \end{array}$$

(2)

$$\begin{array}{ll} \text{a)} & a_k = j \frac{2(-1)^k}{k\pi}, a_0 = 0 \\ \text{b)} & a_k = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2) \\ & b_k = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2) \frac{3}{6 + jk\pi} \\ \text{c)} & a_k = j \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \cos(k\pi/2) \\ & b_k = 2(-1)^k a_k = 2(-1)^k j \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \cos(k\pi/2) \\ \text{d)} & a_k = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2) e^{-jk\pi/3} \\ & b_k = a_k \frac{8\text{sen}(k\pi/4)}{k\pi} e^{-jk\pi/6} = \frac{6}{(k\pi)^2} \text{sen}(k\pi/6) \text{sen}(k\pi/2) e^{-jk\pi/3} \frac{8\text{sen}(k\pi/4)}{k\pi} e^{-jk\pi/6} \end{array}$$

Tema 11: Transformada de Laplace

(1)

- a) $X(s) = \frac{1}{s+2}$ $\Re\{s\} > -2$ es estable
- b) $X(s) = \frac{e^{-2s-6} - e^{-5s-15}}{s+3}$ $\Re\{s\} > -3$ es estable
- c) $X(s) = \frac{2}{s^2} + \frac{1}{s}$ $\Re\{s\} > 0$ no es estable
- d) $X(s) = \frac{4}{s^2+16} + \frac{2}{s^2+16}$ $\Re\{s\} > 0$ no es estable
- e) $X(s) = \frac{1}{(s-2)^2}$ $\Re\{s\} > 2$ no es estable
- f) $X(s) = \frac{2}{s(s^2+4)}$ $\Re\{s\} > 0$ no es estable
- g) $X(s) = -\frac{1}{(s^2+2)}$ $\Re\{s\} > 0$ no es estable
- h) $X(s) = \frac{1-e^{-s}}{s^2} + \frac{1-2e^{-s}}{s}$ $\Re\{s\} > 0$ no es estable
- i) $X(s) = \frac{e^{-2\pi s} - e^{-4\pi s}}{s^2+1}$ $\Re\{s\} > 0$ no es estable
- j) $X(s) = \frac{s^2+2}{s(s^2+4)}$ $\Re\{s\} > 0$ no es estable

(2)

- a) $x(t) = 2e^{-3t}u(t)$
- b) $x(t) = -\frac{j}{14}e^{7jt}u(t) + \frac{j}{14}e^{-7jt}u(t)$
- c) $x(t) = u(t) - 6tu(t) + \frac{5}{5!}t^2u(t)$
- d) $x(t) = \frac{1}{4}\cosh(t)u(t)$
- e) $x(t) = e^{6t}u(t) + 2\cos(\sqrt{6}t)u(t) + \frac{2}{\sqrt{6}}\sen(\sqrt{6}t)u(t)$
- f) $x(t) = u(t) - \frac{1}{2}e^{3t}u(t) + \frac{1}{2}e^{-3t}u(t)$
- g) $x(t) = \frac{1}{4}tu(t) + u(t) - e^{-t}u(t)$

(3)

- a) $y(t) = \left[\frac{1}{5}e^{-2t} - \frac{1}{5}\cos(t) + \frac{2}{5}\sen(t) \right] u(t)$
- b) $y(t) = \left[-\frac{1}{4}t^2 - \frac{1}{2}t - \frac{1}{2} - \frac{1}{2}e^t \right] u(t)$
- c) $y(t) = \frac{5}{6} - \frac{22}{3}e^{3t}u(t) + \frac{13}{2}e^{4t}u(t)$
- d) $y(t) = -e^{2t}u(t) - e^{3t-2}u(t) + 5e^{t+2}u(t)$
- e) $y(t) = \frac{1}{36}e^{3t} + \frac{1}{36}e^{-3t} - \frac{1}{18}\cos(3t)$

(4)

a)

$$h(t) = -\frac{4}{5}e^{-t}u(t) + \frac{9}{10}e^{-2t}\text{sen}(2t)u(t) + \frac{4}{5}e^{-2t}\text{cos}(2t)u(t)$$

b)

$$h(t) = \frac{3}{13}e^t u(t) + \frac{2}{13}e^{-2t}\text{sen}(2t)u(t) - \frac{3}{13}e^{-2t}\text{cos}(2t)u(t)$$

c) No es estable

d) No es Estable

e)

$$h(t) = -\frac{1}{e^4}e^{2t}u(t-2) + 3e^t u(t) + \delta(t)$$

Tema 12: Convolución en Tiempo Discreto

(1)

a)

$$y[n] = \begin{cases} 0 & 0 > n \\ 5\left(\frac{-1}{2}\right)^n \frac{1 - \left(\frac{-2}{3}\right)^{n+1}}{1 + \frac{2}{3}} & n \geq 0 \end{cases}$$

b)

$$y[n] = \begin{cases} 0 & -4 > n \\ -\left(\frac{2}{3}\right)^{n+2} & -4 \leq n \leq -3 \\ -\left(\frac{2}{3}\right)^{n+2} + \frac{27}{4}\left(1 - \left(\frac{2}{3}\right)^{n+3}\right) & n = -2 \\ \frac{27}{4}\left(1 - \left(\frac{2}{3}\right)^{n+3}\right) & -1 \leq n \leq 0 \\ -\frac{27}{4}\left(1 - \left(\frac{2}{3}\right)^n\right) + \frac{19}{4} + 2\left(\frac{2}{3}\right)^{n-3} & 1 \leq n \leq 3 \\ 0 & 3 < n \end{cases}$$

c)

$$y[n] = \begin{cases} 0 & -4 > n \\ n+5 & -4 \leq n < -2 \\ n+6 & -2 \leq n \leq 0 \\ n + \frac{25}{4} & n = 1 \\ -n+8 & 2 \leq n \leq 3 \\ -n+7 & 4 \leq n \leq 6 \\ 0 & 6 < n \end{cases}$$

d)

$$y[n] = \begin{cases} 3^{n+6} - \frac{1}{2}3^{n+1} - \frac{1}{2}3^{n-1} & -6 > n \\ 1 - \frac{1}{2}3^{n+1} - \frac{1}{2}3^{n-1} & -6 \leq n \leq -2 \\ \frac{1}{2}(1 - 3^{n-1}) & -1 \leq n \leq 0 \\ 0 & 0 < n \end{cases}$$

e)

$$y[n] = \begin{cases} 0 & -2 > n \\ 1 & -2 \leq n < 0 \\ 0 & n = 1 \\ n + 2 & 2 \leq n \leq 3 \\ n + 4 & n = 4 \\ 1 & n = 5 \\ -n + 7 & 6 \leq n \leq 7 \\ -1 & n = 8 \\ 0 & 8 < n \end{cases}$$

f)

$$y[n] = \begin{cases} 2^n \frac{10}{9} & 2 > n \\ (\frac{1}{5})^n \frac{1000}{9} & n \geq 2 \end{cases}$$

(2)

a)

$$y[n] = \begin{cases} a^n \frac{1}{1-\frac{1}{a}} & -1 \geq n \\ a^{-1} \frac{1}{1-\frac{1}{a}} & n > -1 \end{cases}$$

b)

$$y_b[n] = y[n-4] = \begin{cases} 2^{n-4} & 3 \geq n \\ 1 & n > 3 \end{cases}$$

c)

$$y_c[n] = \frac{1}{2} y[n-1] = \begin{cases} 2^{n-1} & 0 \geq n \\ \frac{1}{2} & n > 0 \end{cases}$$

d)

$$y_d[n] = y[n] - y[n-10] = \begin{cases} 2 \cdot 2^n - 2 \cdot 2^{n-10} & -1 \geq n \\ 1 - 2 \cdot 2^{n-10} & -1 < n \leq 9 \\ 0 & n > 9 \end{cases}$$

e)

$$y_e[n] = y[n-1] = \begin{cases} 2^{n-1} & n \geq 0 \\ 1 & n < 0 \end{cases}$$